

ddRAD-seq protocol

Natural Resources Institute, University of Greenwich, UK
Vega Lab – Canterbury Christ Church University, UK
Allan Baino

Our protocol combines information from Peterson et al., 2012 and a ddRADseq protocol from the McDaniel Lab – University of Florida, USA. Alterations to bench procedures from our two reference protocols were made to optimize the process for our project, that generated an Illumina sequencing library for SNP discovery in African giant pouched rats (*Cricetomys sp.*) with a ~ 2.5 gb genome.

Like most labs, major steps for ddRADseq library construction remain similar; however, before using this protocol we recommend you familiarize with Peterson et al. (2012), ddRAD experimental design and ddRADseq protocol, and the McDaniel Lab protocol ([McDaniel Lab](#)). Background information about restriction enzyme selection, adapter design and underlying biological theory are not covered in this protocol, and it is assumed that a user is confident with the ddRADseq approach and has previous knowledge on Illumina sequencing.

This protocol utilizes two restriction enzymes: EcoRI-HF and MspI with barcodes on the EcoRI adapters. It is therefore recommended that EcoRI or EcoRI-HF (high fidelity) be one of the restriction enzymes used. This protocol is suitable for the generation of libraries for both single and paired-end sequencing on illumina platforms such as (NextSeq, Hi-Seq and Mi-Seq) and utilizes combinatorial indexing where each individual in a sample in the final sequencing library contains a unique barcode-and-index identifier.

Access to an automated electrophoresis system such as the Agilent Tape Station 4200 that analyses the quantity, size and integrity of samples is highly recommended for potential restriction assay troubleshooting and optimization. Similarly, access to a Pippin Prep or BluePippin electrophoresis system for selecting fragments of desired size distribution compatible with chosen sequencing technology is also highly recommended.

Materials and reagents

EcoRI-HF (cut site G/AATTC supplied with rCutSmart buffer) (NEB, R3101S 20,000 units/ml)

MspI (cut site C/CGG supplied with rCutSmart buffer) (NEB, R0106S 20,000 units/ml)

T4 DNA ligase supplied with buffer (NEB, M0202L 400,000 units/ml)

Adapter P1: annealed oligos 1.1 and 1.2 (barcoded adapters, with EcoRI overhang)

Adapter P2: annealed oligos 2.1 and 2.2 (with MspI overhang)

Annealing buffer stock (10X)

- 100 mM Tris HCl, pH 8
- 500 mM NaCl
- 10 mM EDTA

PCR primers: PCR Primer 1 common to all reactions and Multiplexing/PCR Primer 2 (with unique index) with unique indices (enough to make unique barcode-index combinations for desired sample size). List of oligonucleotides (adapters and primers) can be accessed here: [ddRAD oligonucleotides sheet](#).

100 bp – 600 bp, 2% DF Agarose V2 gel cassettes (Blue Pippin)	
Q5 high fidelity DNA polymerase with Q5 buffer 5X	(NEB, M0491, 2,000 units/ml)
dNTP	(NEB, N0447 8 μ M per dNTP)
AMPure XP beads	(Beckman Coulter Genomics)*
96-well magnetic plate	
Freshly diluted 70% ethanol	
Elution buffer.	(nuclease free water, buffer TE or Qiagen buffer EB)
QIAquick PCR purification kit (250)	(Qiagen, 28106)

** Can opt out if good results are obtained with spin columns (reducing consumables costs) or beads solution prepared in-house.*

Adapter hybridization

In this step, complementary oligonucleotide sequences 1.1 and 1.2 are annealed to form a double stranded Adapter P1 with an overhang that recognizes cut sites on DNA fragments made by restriction enzyme EcoRI-HF. Likewise, oligos 2.1 and 2.2 are annealed to form common Adapter P2 with an overhang that recognizes cut sites made by restriction enzyme MspI.

Complementary oligos for making Adapter P1 and for making common Adapter P2 were combined at a 1:1 ratio in annealing buffer (1X annealing buffer) to make a final annealed adapter stock concentration of 40 μ M from a reconstituted stock at 100 μ M. In our case, we combined 40 μ l of P1.1, 40 μ l of P1.2 and 20 μ l of 1X annealing buffer to make 100 μ l of adapter at 40 μ M. This mixture was then subjected to thermocycling conditions: Pre-heated lid at 50 °C, 1 cycle at 97.5 °C for 2 min and 30 s, 26 cycles at 97.5 °C for 1 min decreasing at 3 °C per cycle to 21 °C, final hold at 4 °C. Final working solutions of these adapters from annealed stock were made and stored at -20 °C.

Double digest restriction assay

It is important to consult product information of chosen restriction enzyme for optimal digest conditions. Typically, in a double digest DNA is digested by both restriction enzymes (in our case EcoRI-HF, which is a six-base DNA cutter, and MspI, which is a four-base DNA cutter). It is critical that enzymes used are compatible for a double-digest assay, i.e. have similar optimal digest conditions such as temperature and use the same reaction buffer. It may be worth using NEBs' Double Digest Finder' ([NEB Cloner](#)) to ensure restriction enzymes are compatible.

We recommend digesting high quality genomic DNA between 100 – 500 ng in a 20-50 μ l reaction volume; however, we have had good success digesting between 250 – 500 ng of DNA. The 'Gold Standard' is to digest DNA at concentrations of at least 20 ng/ μ l; however, this is not always possible due to differences in extraction methods and natural degradation acting on a sample (in cases of long-term storage). We recommend starting with similar amounts of DNA regardless of concentration, this normalization is important because the number of loci and reads generated from sequencing are highly correlated with the quantity of DNA in samples. Uneven quantities of initial DNA can result in uneven number of loci and reads between samples. Quantifying the amount of dsDNA with a Qubit fluorometer in samples is

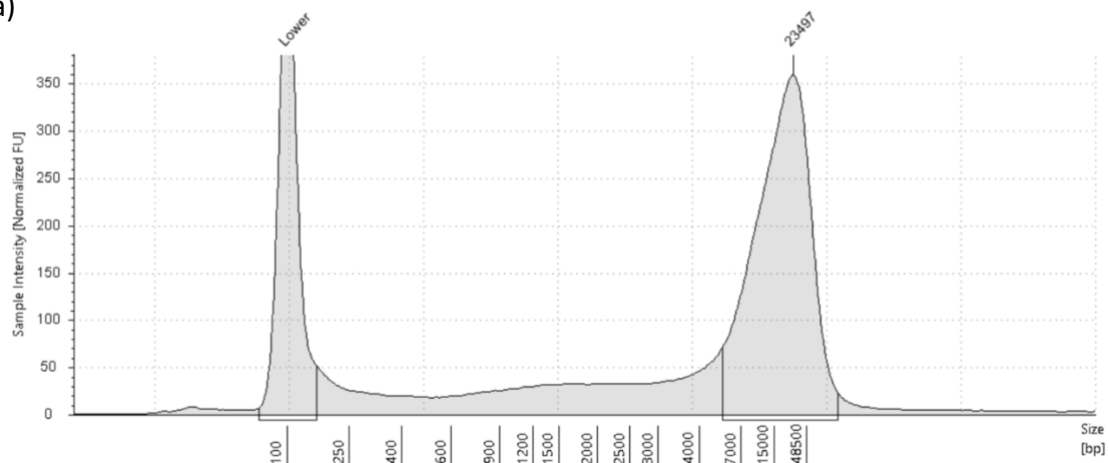
highly recommended to establish how much volume of DNA should be added to the total digest reaction volume.

Run the double digestion reaction as appropriate for the chosen restriction enzymes and amount of DNA in reaction volume. To allow for complete digestion, we added equal volumes of both EcoRI-HF and MspI restriction enzymes (REs) whose volume amounted to 8% of the total reaction volume. Thermocycling conditions for digest reaction were: heated lid at 50 °C, incubation for 12 hours at 37 °C, cool reaction to room temperature, final hold at 4 °C. Do not heat-inactivate enzymes as this may skew base composition in final library.

We recommend quantifying digests before and after clean-up to have an idea on how much dsDNA is available for the next phase of library prep (adapter ligation). Simple Qubit fluorometric quantification can provide reliable readings; however, we recommend running samples on an Agilent Tape Station 4200 immediately after digest to visualize if desired size distribution of DNA fragments is present in the digested samples (Figure 1 a-d).

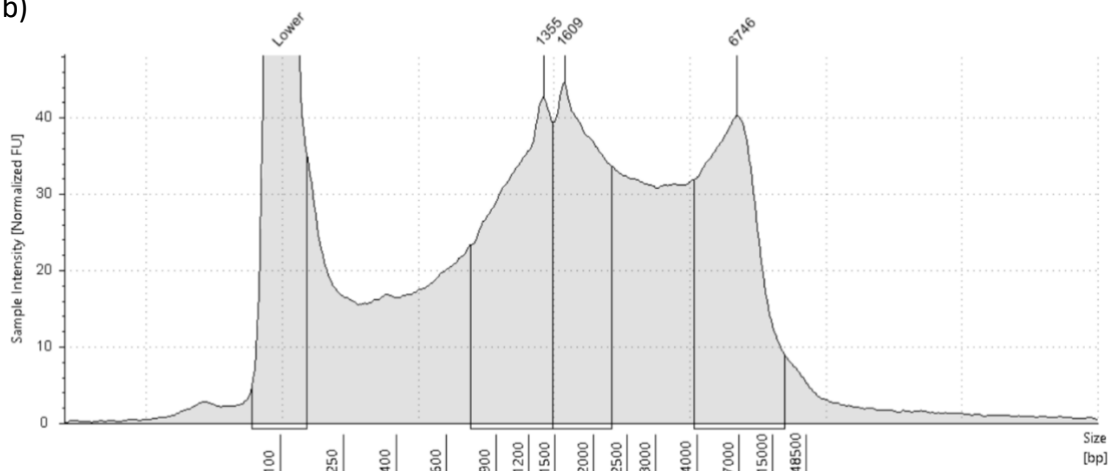
B1: C096und

(a)

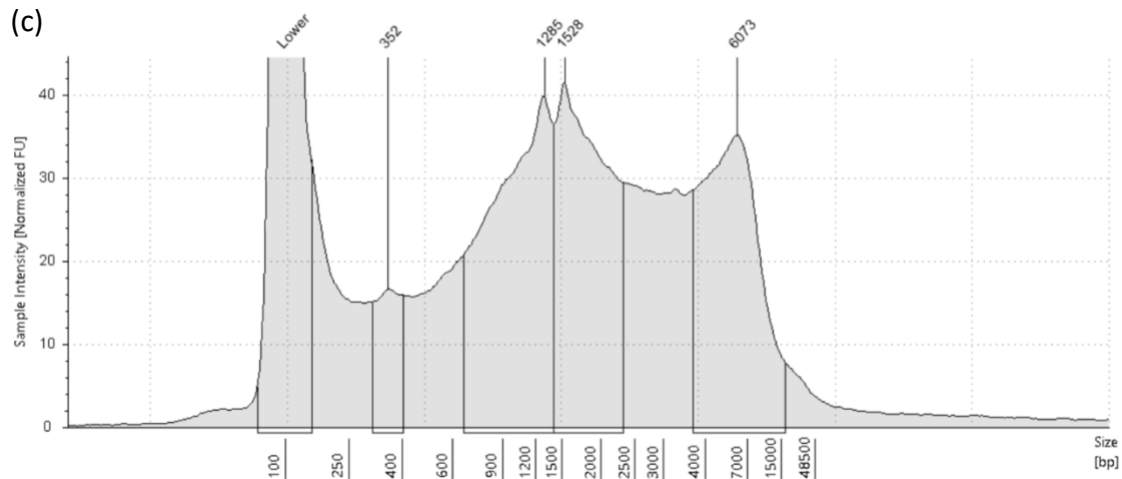


C1: C096_5%

(b)



D1: C096_8%



E1: C096_10%

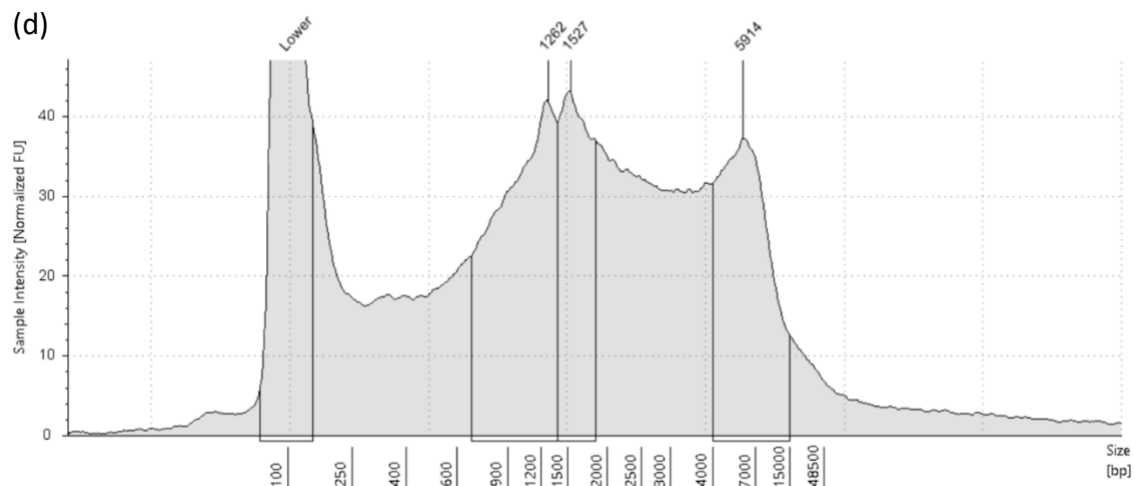


Figure 1: (a) Undigested DNA sample, (b) digested DNA sample showing a molecular weight shift towards lower fragment size distribution when DNA sample in is subjected to double digest restriction enzyme (REs) assay with REs at 5% of the total reaction volume, (c) slightly greater reduction in high molecular weight DNA towards lower fragment size range when REs added to digest assay are at 8% of the total reaction volume, and (d) similar size distribution of DNA fragments as appears in (b) and (c) with REs at 10% of total reaction volume in a digest assay.

Adapter ligation and check

At this juncture we want to ligate double stranded adapters onto overhangs or sticky ends from EcoRI-HF and MspI cut sites. EcoRI-HF adapters contain a unique barcode sequence that serves as a sample identifier while adapter P2 (MspI) common to all samples contains a forked portion in its sequence that permits incorporation of flow cell annealing oligonucleotides from the P1 end during initial cycle of i7 indexing (Figure 2).

Oligos; Adapters; Digested genomic DNA

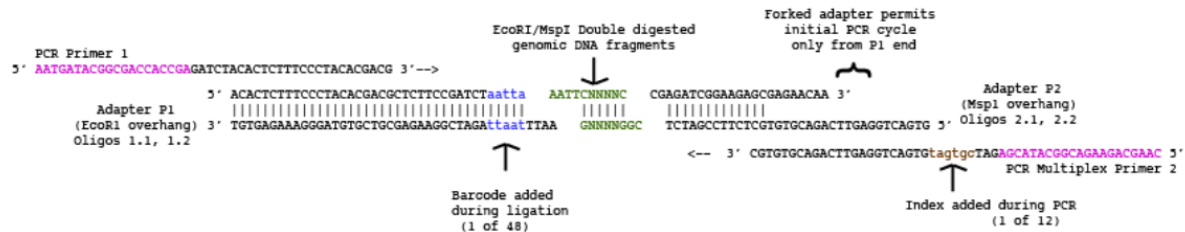


Figure 2: Extract from Peterson et al. (2012) depicting adapter ligation process (adapter P1 with EcoRI overhang and Adapter P2 with MspI overhang) and final indexing PCR (with PCR Primer 1 and PCR Multiplex Primer 2) that completes sequencing library construction. The resulting DNA fragment will contain the unique combinatorial adapter-index for each individual.

Ligation efficacy depends on adapter to digested DNA fragment ends ratio, a 2 to 10 fold adapter to DNA fragment sticky ends molar excess is recommended for efficient ligations. We opted for an 8-fold molar excess using Peterson et al. (2012) ligation molarity calculator spreadsheet available online ([ligation molarity calculator](#)). For the ligations, a mix of 200 µl of double stranded adapters P1 and P2 working solutions were prepared (one working solution for each barcoded adapter). In a 40 µl ligation reaction, the P1 and P2 adapters working solution and digested DNA were added at a 1:1 ratio, with 2 µl of T4 ligase and 4 µl of T4 ligase buffer and nuclease free water to make 40 µl reaction volume. Thermocycling conditions for ligation were: heated lid at 50 °C, incubation for 12 hours at 16 °C, heat inactivation at 65 °C, final hold at 4 °C.

Ligation check by PCR

At this juncture, it is important to validate the ligation reactions by testing successful library construction for each DNA sample. One simple way of doing this is by PCR and checking if there is amplification; a successful PCR would indicate that the ligation step worked, and downstream steps can then proceed. To validate the ligation reactions, a 15 µl PCR reaction was performed with inexpensive polymerase (2X Dream Taq polymerase) and with PCR Primer 1 and Multiplex PCR Primer 2 (which are the ones that contain the Illumina flow cell binding sequences and permit integration of these sequences onto the ends of the ligated DNA sample). In doing so, this PCR reaction verifies whether both adapters (P1 and P2) successfully ligated to respective ends of the DNA fragment. Thermocycling conditions for PCR check were: heated lid at 105 °C, 95 °C for 5 min, 20 cycles (95 °C for 30 s, 52 °C for 40 s, 72 °C for 1.30 min), final extension at 72 °C for 7 min, and final hold at 4 °C. PCR products for each sample were ran through a 1.4% Agarose gel with 100 bp ladder; samples with a successful ligation had a bright smeared lane of fragments ~ 100 bp to >700 bp while samples with a failed ligation had faint smears ranging 50 bp to 200 bp (adapters and primer dimers) (Figure 3).

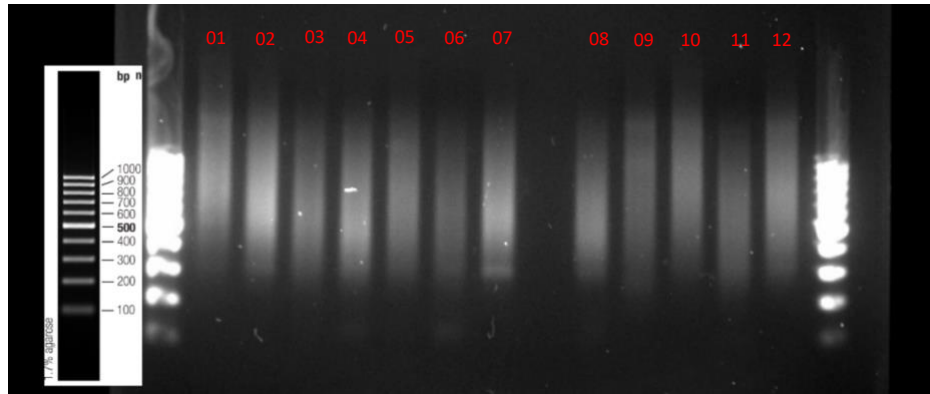


Figure 3: Gel image validating ligation success by PCR. The first and last gel lanes are our 100bp ladder while smeared lanes depict successful library construction for individual DNA samples. In this case, there were no failed ligations.

Pooling of barcoded samples for size selection

At this stage, uniquely barcoded samples can be pooled by respective multiplexing primer with a unique index. In this case, a combinatorial barcode-and-index that produces a unique combination of barcodes in adapter P1 and one of the 12 multiplex indices. This results in a unique dual-index for each individual specimen (Figure 4). Besides from uniquely identifying an individual sample, combinatorial indexing also facilitates bioinformatic demultiplexing of individuals after Illumina sequencing.

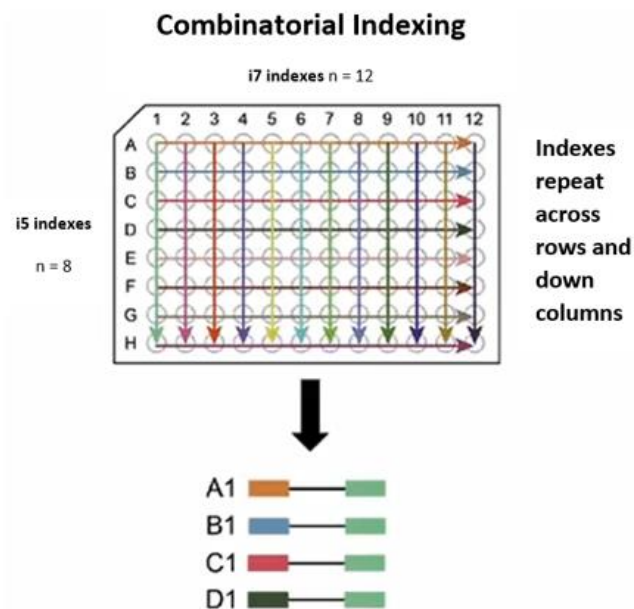


Figure 4: Use of unique barcodes in P1 adapter and unique index in PCR 2 multiplexing primers. Barcodes/index i5 sequences (introduced during ligation with adapter P1) repeat across rows and index i7 (introduced with multiplexing PCR 2 primer) down columns.

We took 10 μ l of ligated DNA from each individual sample (that produced successful library construction during the ligation PCR check) into respective PCR2 multiplexing pools (12 pools total; e.g., individual 1 with barcode 1 and index 1, individual 2 with barcode 2 and index 1, individual 3 with barcode 3 and index 1, etc). Sample pools were quantified for dsDNA concentration on a Qubit fluorometer 2.0 (dsDNA BR assay) and then cleaned using spin columns to remove inactivated ligation enzyme, excess adapter and buffer, and to concentrate the sample as only a small volume needs to be loaded onto the Blue Pippin for size selection. We performed size selection using 100 bp – 600 bp, 2% DF Marker V2 Agarose gel cassettes with an internal reference marker. Approximately 10 μ l of internal reference marker and 30 μ l of each sample pool were prepared in triplicate and loaded onto the Blue Pippin to select DNA fragment distributions of ranges: 250 bp – 400 bp, 400 bp – 550 bp and 550 bp – 650 bp. Eluted DNA was then visualized on an Agilent Tape Station 4200 to validate selected size distribution (Figure 5). Size distribution 250 bp – 400 bp had the highest density across all 12 elution pools. It is important to note that ~ 76 bp of adapter sequence were added during ligation; subtraction provides actual gDNA insert size.

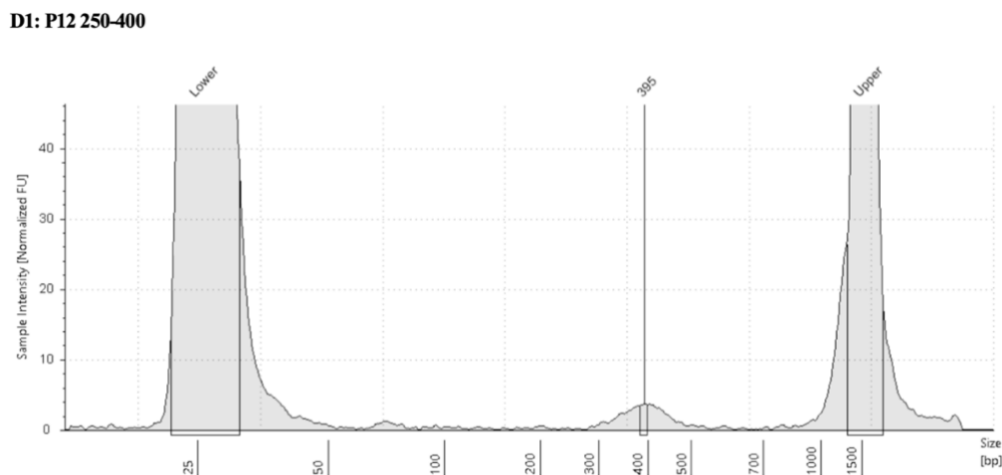


Figure 5: Tape station output showing the size distribution of ligated DNA in a sample of 9 pooled individuals after a BluePippin Prep 250 bp – 400 bp size selection run.

i7 indexing PCR

For each of our 12 size-selected sample pool elutions, we added the Illumina flow cell annealing multiplexing primer sequence (required for index reads, paired-end sequencing). To do this, we performed an indexing PCR with a high fidelity polymerase (Q5 DNA polymerase) to increase concentrations of each library pools before final equimolar pool. Our selection of 11 barcodes and 12 PCR2 multiplexing or i7 indexing primers (combinatorial indexing) allows for pooling and sequencing of up to 132 individual samples in a single lane of sequencing. For each Blue Pippin elution pool, we prepared four to eight aliquots with: (3 μ l) of size selected ligated DNA, PCR primers 1 and 2 (5 μ M each), Q5 buffer (5X), dNTPs, Q5 Taq polymerase and nuclease free water in a 20 μ l PCR reaction volume. Thermocycling conditions were: heated lid at 105 $^{\circ}$ C, 98 $^{\circ}$ C for 30 s, 15 - 20 cycles (98 $^{\circ}$ C for 15 s, 60 $^{\circ}$ C for 30 s, 72 $^{\circ}$ C for 30 s), final extension at 72 $^{\circ}$ C for 2 min, and final hold at 4 $^{\circ}$ C. The PCR products were pooled and imaged

on a 1.4% Agarose gel. Successful amplifications were quantified on a Qubit 2.0 fluorometer (dsDNA HS assay kit), cleaned using QIAquick (Qiagen) spin columns (by pool). Cleaned libraries (12 pools) were eluted at 50 µl into 1.5 ml Eppendorf tubes, equimolar amounts from each pool were combined to make a final library of 100 µl. The final library comprising 108 samples (single sample depicted in figure 6) was visualized on an Agilent Tape station 4200 before sending to a sequencing facility.

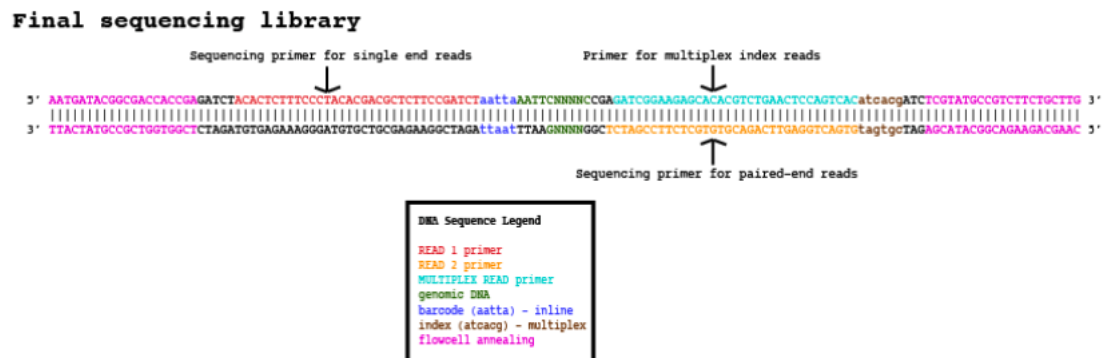


Figure 6. Extract from Peterson et al. (2012) depicting a single dual-indexed sample within the final sequencing library.